Moller scattering 3

From the first section we have

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* = \frac{e^4}{t^2} \left(8s^2 + 8u^2 - 64m^2 s - 64m^2 u + 192m^4 \right)$$

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* = \frac{e^4}{tu} \left(-8s^2 + 64m^2 s - 96m^4 \right)$$

$$\sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* = \frac{e^4}{u^2} \left(8s^2 + 8t^2 - 64m^2 s - 64m^2 t + 192m^4 \right)$$

We also have by the Casimir trick

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* = \frac{e^4}{t^2} \operatorname{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \operatorname{Tr} \left((\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right)$$

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* = \frac{e^4}{tu} \operatorname{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu (\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right)$$

$$\sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* = \frac{e^4}{u^2} \operatorname{Tr} \left((\not{p}_3 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \operatorname{Tr} \left((\not{p}_4 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right)$$

The trace operator Tr contracts (sums over) the first and last index. Slash matrices are formed as

$$\not{p} = p^\mu g_{\mu\nu} \gamma^\nu = p^0 \gamma^0 - p^1 \gamma^1 - p^2 \gamma^2 - p^3 \gamma^3$$

Eigenmath script